PROJECT MODULE: TASK 3 CONFIGURATION REPORT

Target Infrastructure: Ubuntu 24.04 LTS Instance on Microsoft Azure

1. Scope & System Objective

Implement a high-frequency background data collector shell mechanism to monitor isolated runtime metrics (CPU load metrics and live memory allocations). This logs performance stats continuously with high-precision timestamp tracking indicators.

2. High-Frequency Monitor Implementation Layout

- **Asset A: Automated Core Analytics Engine Script (monitor.sh)** :

Bash

```
#!/bin/bash

LOG_FILE="/opt/container-monitor/logs/container_stats.log"

TIMESTAMP=$(date "+%Y-%m-%d %H:%M:%S")


# Format output logic explicit checks updates structure text line mapping rules check syntax verification

CONTAINER_STATS=$(docker stats my-running-app --no-stream --format "Container: {{.Name}} | CPU: {{.CPUPerc}} | Mem: {{.MemUsage}}")


echo "[$TIMESTAMP] $CONTAINER_STATS" >> $LOG_FILE
```

- **Asset B: Script Executable Flag Modification** :

Bash

```
chmod +x /home/monitor_user/monitor.sh
```

3. Step-by-Step Scheduling Integration

Accessed cron process orchestration matrices under the dedicated identity space:

Bash

```
crontab -e
```

Injected the automation cron rule parameters explicitly at the bottom line of the configuration map:

Plaintext

```
* * * * * /bin/bash /home/monitor_user/monitor.sh
```

4. Verification & Validation Metrics

Querying data updates patterns inside the log directory path demonstrates continuous one-minute background processing traces:

Plaintext

$ cat /opt/container-monitor/logs/container_stats.log

[2026-06-18 15:21:01] Container: my-running-app | CPU: 0.00% | Mem: 2.633MiB / 7.756GiB

[2026-06-18 15:22:01] Container: my-running-app | CPU: 0.01% | Mem: 2.635MiB / 7.756GiB